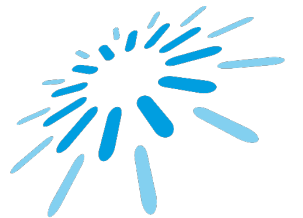




Collaboration, Quality, Test

Prof. Dr. Jóakim von Kistowski



TH Aschaffenburg
university of applied sciences

Choose your next Prof!

**Wednesday, Nov. 19
and Thursday, Nov. 20**
Campus 1 - Room 26/214

**No CQT lectures and exercises on
those days!**

	Monday	Tuesday	Wednesday	Thursday	Friday
08:00 - 09:30					
09:45 - 11:15				Vorlesung Software Design	
11:30 - 13:00			Lecture Software Design International	Übung Software Design	
	Lunch Break				
14:00 - 15:30			Exercises (Group 2) Software Design International	Exercises (Group 1) Software Design International	
15:45 - 17:15			No lectures and exercises on Nov. 19 and 20!		
17:30 - 19:00					

Check-In, Part I

Recap:
Think/Remember an example case:
Why is testing necessary?

Recap:
Think/Remember an example case:
Why is ***automated*** testing necessary?

Some thoughts on complete tests ...

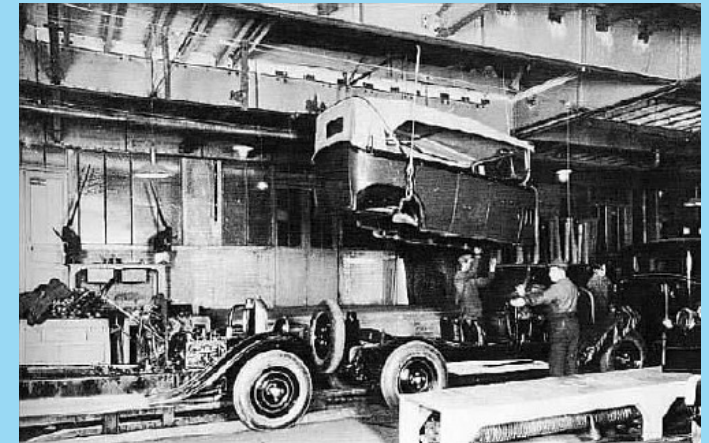
**Strike the following sentence from
your vocabulary:**

**„We have completely
tested everything / all cases“**

How do others test?

Automotive industry

- Incoming test of the components (component test)
- Ongoing intermediate checks on the assembly line (integration test)
- Test in use under (almost) real-world conditions (system test)
- Customer test drive (acceptance test)
- Race use (performance test, load test)
(stability, reliability, robustness)
- Crash test (stress test)



Is complete testing possible?

- Given a simple program
 - three integer input values
- Data type of input values: 16-bit integers
 - $\rightarrow 2^{16}$ different values possible
- three independent input values:
 - $\rightarrow 2^{16} * 2^{16} * 2^{16} = 2^{48}$ combinations.
- For a complete test: Each of these combinations must be tested.
- How long does it take with 100,000 test cases per second?

281.474.976.710.656 test cases
Duration: about 90 years

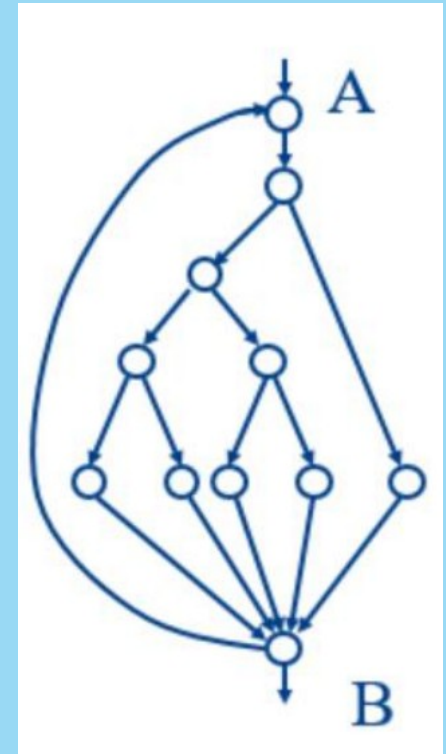
Is complete testing possible?

- To be tested: A simple program that consists of four branches (IF statements) and a comprehensive loop, thus containing five possible paths in the loop body.
- Assuming the branches are independent of each other and limiting the number of loop passes to a maximum of 20, the following calculation results:

$$5^1 + 5^2 + \dots + 5^{18} + 5^{19} + 5^{20}$$

- To be tested: All paths with a maximum of 20 loop passes
- How long does it take with 100,000 test cases per second?

119.209.289.550.780 test cases
Duration: about 38 years



Is complete testing possible?

To be tested: A program that reads in 3 positive integer values and interprets them as the length of a triangle.

The program outputs a message as to whether it is an **unequal** (all sides different), **isosceles** (two sides equal), or **equilateral** (all sides equal) triangle.



$$a \neq b \neq c$$



$$\begin{aligned} a &= b \neq c \\ a &\neq b = c \\ a &= c \neq b \end{aligned}$$



$$a = b = c$$

Your suggestions

How many tests are needed?

What would you test?

Possible Test cases (1/3)

Test cases consist of test data and the expected result

1. 2,3,4 – valid unequal triangle
2. 2,2,2 – valid equilateral triangle
3. 2,2,1 – valid isosceles triangle
- 4./5. 1,2,2 / 2,1,2 – permutations for isosceles triangles
6. 1,0,3 – no triangle, one side = 0
- 7./8. 0,1,3 / 1,3,0 – permutations
9. 5,-5,9 – no triangle, one side < 0
- 10./11. -5,5,9 / 5,9,-5 – permutations

Possible Test cases (2/3)

- 12.** 1,2,3 – no triangle
sum of shorter sides = 3rd side
- 13/14.** 2,3,1 / 3,1,2 – permutations
- 15.** 1,2,4 – no triangle
- 16./17.** 2,4,1 / 4,1,2 – permutations
- 18./19.** 0,0,0 – no triangle or error, all sides = 0
additionally, test if 2 sides = 0
- 20. – 22.** Max_int, Max_int, Max_int – valid equilateral triangle
additionally, test with 2 or 1 max value

Possible Test cases (1/3)

- 23.-25.** 1,1,4.4567 – Error „only integer numbers“
test permutations?
additionally, test with 2 or 3 floating point values
- 26.-28.** 1,1,& – Error „only integer numbers, no letters or special characters“
test permutations?
additionally, test with 2 or 3 letter / special character values
- 29./30.** 1,2,3,4 / 2,3 – Error „unexpected number of input parameters“ (if possible)
- 31.** $\text{Max_int}/2 + 1$, $\text{Max_int}/2 + 1$, $\text{Max_int}/2 + 10$ – valid isosceles triangle (test for overflow)

How many test cases did you come up with?

What does this mean for you?

Conclusion:

Easy problem, difficult test!

The Test ...

- determines **speed of development**. Untested things cannot be delivered.
- gets more and more **complex** (as programs become more complex).
- gets more and more **important**. The more relevant the software (e.g. safety-critical), the more important the test.
- is **far from complete**. We need “clever test strategies”.
- is much more than just “clicking around a little”. Systematic testing can be learned and is part of professional development.
- needs **experts!** (and qualifications ;-)

Learning Objectives for Fundamentals of Testing

(according to Certified Tester Foundation Level Syllabus, English Version V4.0)

- **1.1 What is Testing?**
- FL-1.1.1 (K1) Identify typical test objectives
- FL-1.1.2 (K2) Differentiate testing from debugging
- **1.2 Why is Testing Necessary?**
- FL-1.2.1 (K2) Exemplify why testing is necessary
- FL-1.2.2 (K1) Recall the relation between testing and quality assurance
- FL-1.2.3 (K2) Distinguish between root cause, error, defect, and failure

Learning Objectives for Fundamentals of Testing

(according to Certified Tester Foundation Level Syllabus, English Version V4.0)

- **1.3 Testing Principles**
- FL-1.3.1 (K2) Explain the seven testing principles
- **1.4 Test Activities, Testware and Test Roles**
- FL-1.4.1 (K2) Summarize the different test activities and tasks
- FL-1.4.2 (K2) Explain the impact of context on the test process
- FL-1.4.3 (K2) Differentiate the testware that supports the test activities
- FL-1.4.4 (K2) Explain the value of maintaining traceability
- FL-1.4.5 (K2) Compare the different roles in testing
- **1.5 Essential Skills and Good Practices in Testing**
- FL-1.5.1 (K2) Give examples of the generic skills required for testing
- FL-1.5.2 (K1) Recall the advantages of the whole team approach
- FL-1.5.3 (K2) Distinguish the benefits and drawbacks of independence of testing

Chapter 1: Fundamentals of Testing



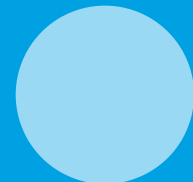
Terms and Motivation

Testing Principles

Test Activities, Testware and Test Roles

Test Cases, Expected Results, and Test Oracle

Essential Skills



What is an error, what is a defect?

Remember:

**A situation can only be
classified as an error if the
expected/correct has been
specified.**

The Chain of Causes and Errors

- Error/Mistake
 - produced by human (or AI) when creating the program/application
 - Its effect is observed when executing the software
- Failure: The observable instance of the error occurring in execution
- Defect: The failure's cause (e.g., in code)
- Root Cause: The fundamental reason for the occurrence of a problem

Task

Draw a graphical representation (diagram) showing how errors, failures, defects, and root causes relate to each other.

Try not to google it first

Failure – Definition

- Event in which a component or system does not perform a required function within the specified scope. [after ISO 24765]
- Effect of a defect that is visible to the “outside world”.
- Deviation between expected result and actual result.
- Deviation of a system from the expected delivery/performance.

Notes on the effect of failures

- (rarely) caused by cosmic radiation, electromagnetic fields or hardware errors.
- Can lead to incorrect output, slow execution or abort.

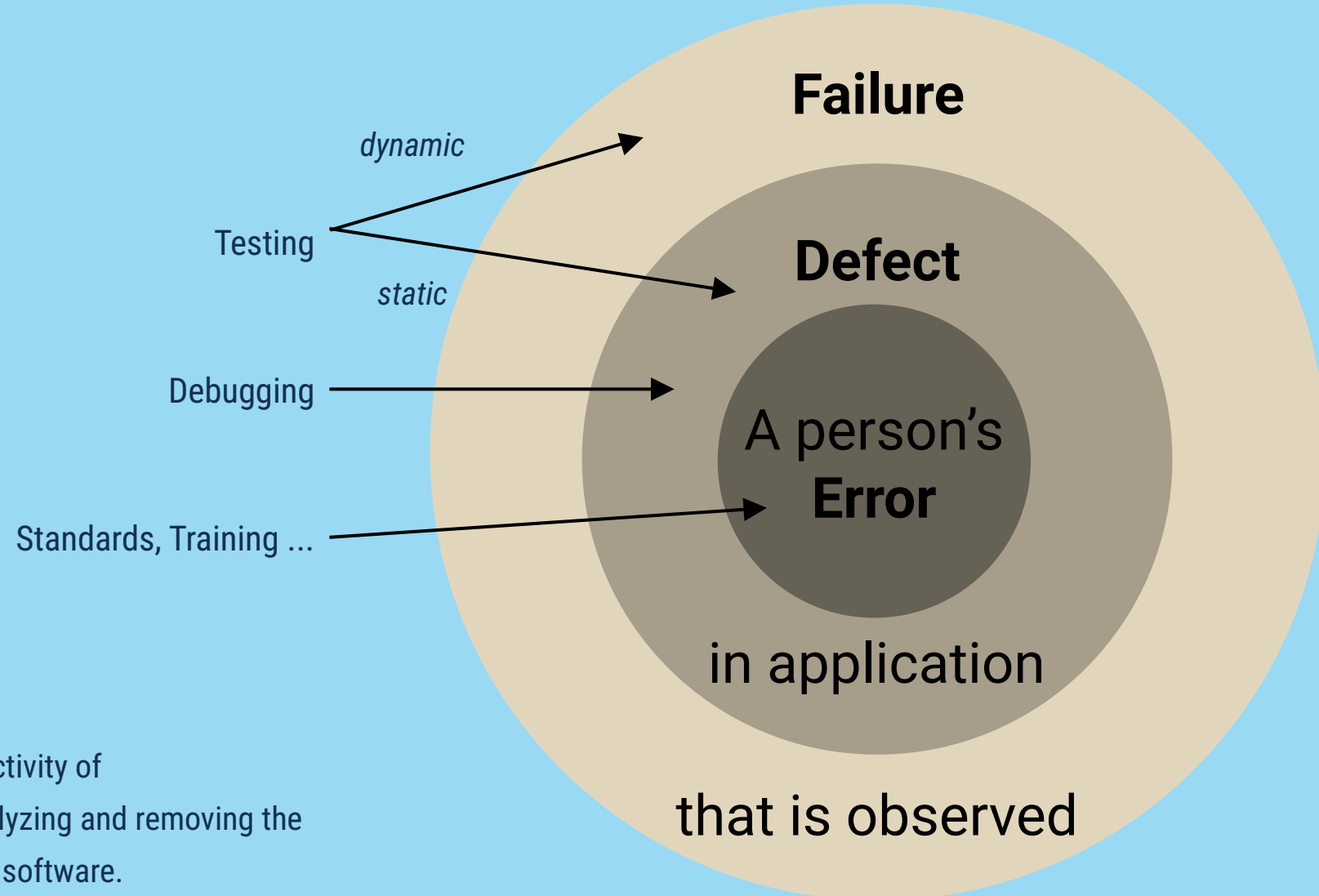
Defect – Definition

- Inadequacy or defect in a work product so that it does not meet its requirements or specifications [after ISO 24765]
- Incorrect software part
 - Incorrect instruction
 - Incorrect data definition
- Condition of a program that may (e.g. high load) impair a required function or lead to errors

Error – Definition

- The human action that leads to an incorrect result [after ISO 24765]
- The human action
 - Developer: leads to a defect in the software
 - User: incorrect operation with undesirable result
- Unknowingly, inadvertently or intentionally performed action or omission that may lead to impairment of the software function

Relationships: Error, Defect, Failure



*Note: Debugging is an activity of locating/identifying, analyzing and removing the causes of failures in the software.

Testing – Definition (after ISO 24765)

Process consisting of all **static and dynamic** life cycle activities relating to

- **Planning**
- **Preparation**
- **Evaluation**

of a component or system and associated work results.

Process goal: Determine

- If they meet the specified requirements
- If they are suitable for the given purpose
- If they are error free.

Testing – Definition (after ISO 24765)

Process consisting of all **static and dynamic** life cycle activities relating to

- **Planning**
- **Preparation**
- **Evaluation**

of a component or system and associated work results.

What does this mean?

Process goal: Determine

- If they meet the specified requirements
- If they are suitable for the given purpose
- If they are error free.

What are test objectives?

Test Objectives

Potential objectives when testing

- Uncovering defects and faults
- Evaluate work results: requirements, user stories, architecture design and code
- Identify errors in the work results and avoid them in subsequent work results
- Verify whether all specific requirements have been met
- Check completeness of the test object
- Validate the test object against user and other stakeholder expectations
- Create confidence in the quality level of the test object
- Reduce risks due to inadequate software quality
- Provide stakeholders with information to make informed decisions, especially regarding the quality level of the test object

Validation vs. Verification (1/6)

Which is which?

Test
"against
requirements"

Test
"against
application
context"

Validation vs. Verification (2/6)

Which is which?

Are we doing
the right thing?

Are we doing
it right?

Validation vs. Verification (3/6)

Which is which? – **Requirements:**

Made of plastic,
blue

Vessel to be
filled with water

Vessel to be
filled with water

With handle

Validation vs. Verification (4/6)

Which is which? – **Requirements:**



Requirements met?

Verification ✓

Made of plastic,
blue

Vessel to be
filled with water

Vessel to be
filled with water

With handle

Validation vs. Verification (5/6)

Which is which? – **Requirements:**



Usable?

Validation **X**

Made of plastic,
blue

Vessel to be
filled with water

Vessel to be
filled with water

With handle

Validation vs. Verification

Which is which?

Validation

- Confirmation by providing objective evidence that the requirements for a specific **intended use** or application have been met.
- Did we implement **the right system**?

Verification

- Confirmation by providing objective evidence that specified **requirements have been met**.
- Did we implement the **system the right way**?

Chapter 1: Fundamentals of Testing

Quality



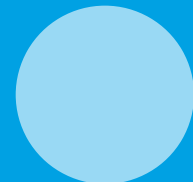
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Essential Skills



What is Quality? (1/4)

- Degree to which the system satisfies the **stated** and **implied** needs of its **various stakeholders** [ISO 25010].
- Degree to which a set of inherent characteristics fulfills requirements [after ISO 9000:2000]

What is Quality? (2/4)

- Degree to which the system satisfies the **stated** and **implied** needs of its **various stakeholders** [ISO 25010].
- Degree to which a set of inherent characteristics fulfills requirements [after ISO 9000:2000]
- You want to buy a car:
 - Which requirements would you mention explicitly?
 - Which requirements would you take for granted?

What is Quality? (3/4)

- Degree to which the system satisfies the **stated** and **implied** needs of its **various stakeholders** [ISO 25010].
- Degree to which a set of inherent characteristics fulfills requirements [after ISO 9000:2000]
- You want to buy a car:
 - Which requirements would you mention explicitly?
 - Which requirements would you take for granted?
- Example Chat App:
 - Which requirements would you mention explicitly?
 - Which requirements would you take for granted?

What is Quality? (4/4)

- Degree to which the system satisfies the **stated** and **implied** needs of its **various stakeholders** [ISO 25010].
- Degree to which a set of inherent characteristics fulfills requirements [after ISO 9000:2000]
- Aspects of quality relate to requirements
 - Functional requirements
(e.g., Domain specifics, Correctness, Suitability)
 - Non-functional requirements
(e.g., quality requirements, performance, IT-security, usability, ...)

ISO 25010

The standard on software quality



Quality according to ISO 25010

Select three apps on your smartphone

- a chat app
- an app you use daily
- an app you use rarely

Task 1.1: For each app, rate the importance of the respective quality aspects according to ISO. Look up the definitions for the respective quality characteristics in the ISTQB® Glossary.

Task 1.2: Do the same thing for your vehicle (Car, Bike, Train, etc.).



Functional
Suitability



Reliability



Usability



Security



Compatibility



Portability



Maintainability



Performance
Efficiency



Quality according to ISO 25010

Select three apps on your smartphone

- a chat app
- an app you use daily
- an app you use rarely

Task 1.3: Prioritize the quality aspects.
Sort them **uniquely** according to their
importance for each application.



Functional
Suitability



Reliability



Usability



Security



Compatibility



Portability



Maintainability



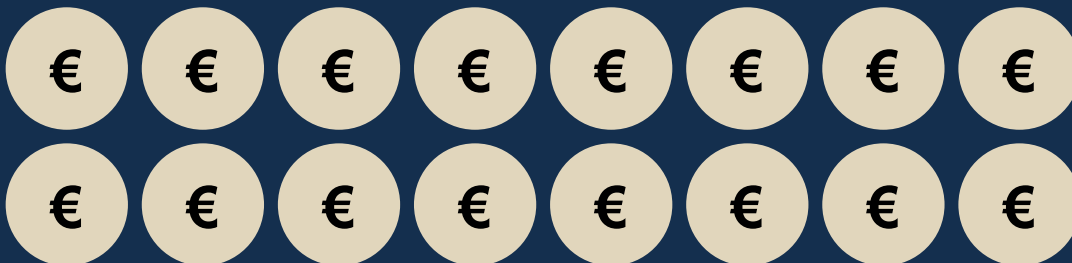
Performance
Efficiency



Quality according to ISO 25010

**Task 1.4: You have a budget of 16
“Quality-Points” (1 Point = 10k €) per
app. Distribute the points!**

**Prioritize the quality aspects and
invest accordingly.**



Functional Suitability	★ ★ ★ ★ ★
Reliability	★ ★ ★ ★ ★
Usability	★ ★ ★ ★ ★
Security	★ ★ ★ ★ ★
Compatibility	★ ★ ★ ★ ★
Portability	★ ★ ★ ★ ★
Maintainability	★ ★ ★ ★ ★
Performance Efficiency	★ ★ ★ ★ ★

How expensive is your Chat app?

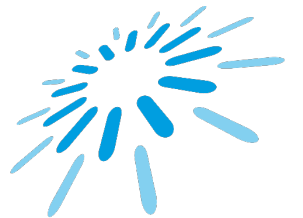
You should now be able to answer these questions

- Define the terms: Error, Fault, Failure
- Explain the difference between testing and debugging.
- Define the terms verification and validation.
- Explain why every test is a random sample test.
- Name the main characteristics of software quality according to ISO.
- Define the term reliability of a system.



Until next time!

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